

DSA PROBLEM NOTES

Two Sum • LeetCode #1

1. PROBLEM IN MY WORDS

Find two numbers whose sum equals the target.

2. EXAMPLE

Input	nums = [2, 7, 11, 15], target = 9
Answer	[0, 1]

3. BRUTE FORCE

Check every possible pair using two loops.

Time: $O(n^2)$

Space: $O(1)$

4. PATTERN

HashMap

Why?

We need a fast lookup for $\text{target} - \text{current}$.

5. CORE IDEA

For every number:

$\text{needed} = \text{target} - \text{current}$

If needed already exists $\rightarrow$ answer.

Otherwise $\rightarrow$ store the current number.

6. DRY RUN

nums = [2, 7, 11, 15], target = 9

$i = 0 \rightarrow \text{current} = 2 \rightarrow \text{needed} = 7 \rightarrow 7$ not found $\rightarrow$ store 2

$i = 1 \rightarrow \text{current} = 7 \rightarrow \text{needed} = 2 \rightarrow 2$ found $\rightarrow$ answer [0, 1]

7. CODE

```
class Solution {
    public int[] twoSum(int[] nums, int target) {
        Map<Integer, Integer> map = new HashMap<>();
        for (int i = 0; i < nums.length; i++) {
            int needed = target - nums[i];
            if (map.containsKey(needed))
                return new int[]{map.get(needed), i};
            map.put(nums[i], i);
        }
        return new int[]{};
    }
}
```

8. COMPLEXITY

Time Complexity: $O(n)$

Space Complexity: $O(n)$

9. MISTAKES ⚠

- Don't use the same element twice.
- Check the HashMap before inserting the current element.

10. RECOGNITION CLUE 🔑

Pair + target + fast lookup → HashMap

11. RELATED PROBLEMS

- 3Sum
- 4Sum
- Subarray Sum Equals K